

Uvod

Ocenjujemo neznanu konstantno količino x

$$\begin{cases} z_1 = x + r_1 \\ z_2 = x + r_2 \end{cases} \quad \langle r_1^2 \rangle = \sigma_1^2, \langle r_1 r_2 \rangle = \sigma_{12}$$

Želimo združiti z_1 in z_2 v novo oceno za x :

$$\hat{x} = x + \tilde{r}$$

$$\text{pogoj: } \langle \tilde{r} \rangle = 0$$

$$\hat{x} = a z_1 + b z_2$$

$$x + \tilde{r} = (a+b)x + a r_1 + b r_2 \quad / \cdot \langle \rangle$$

$$x = (a+b)x \rightarrow a+b=1$$

$$\hat{x} = (1-b)z_1 + b z_2 = z_1 - b \underbrace{(z_2 - z_1)}_{\substack{\text{interakcija} \\ \text{faktor}}}$$

→ iščemo optimalno b_0 , ki nam da optimalno oceno:

$$\hat{x} = \hat{x} \\ \downarrow \\ \text{optimalna ocena}$$

$$\rightarrow \text{zahtevamo min. varianco: } \frac{d\sigma^2}{db} = 0 \rightarrow b_0$$

$$\sigma^2 = \langle \tilde{r}^2 \rangle = \langle ((1-b)r_1 + b r_2)^2 \rangle = \langle (1-b)^2 r_1^2 + b^2 r_2^2 + 2(1-b)b r_1 r_2 \rangle =$$

$$= (1-b)^2 \sigma_1^2 + b^2 \sigma_2^2 + 2(1-b)b \sigma_{12}$$

$$\frac{d\sigma^2}{db} = -2(1-b)\sigma_1^2 + 2b\sigma_2^2 + 2(1-2b)\sigma_{12} = 0$$

$$b_0 (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) - \sigma_1^2 + \sigma_{12} = 0$$

$$b_0 = \frac{\sigma_1^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

$$\rightarrow \hat{x} = z_1 + \frac{\sigma_1^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} (z_2 - z_1)$$

$$\sigma^2 = (1-b_0)^2 \sigma_1^2 + b_0^2 \sigma_2^2 + 2(1-b_0)b_0 \sigma_{12} =$$

$$= \frac{1}{I^2} \left[(\sigma_1^2 - \sigma_{12})^2 \sigma_1^2 + (\sigma_1^2 - \sigma_{12})^2 \sigma_2^2 + 2(\sigma_1^2 - \sigma_{12})(\sigma_1^2 - \sigma_{12}) \sigma_{12} \right] =$$

$$= \frac{1}{I^2} \left[(\sigma_1^2 - \sigma_{12}) \left\{ (\sigma_1^2 - \sigma_{12}) \sigma_1^2 + (\sigma_1^2 - \sigma_{12}) \sigma_2^2 \right\} + (\sigma_1^2 - \sigma_{12}) \left\{ (\sigma_1^2 - \sigma_{12}) \sigma_2^2 + (\sigma_2^2 - \sigma_{12}) \sigma_{12} \right\} \right] =$$

$$= \frac{1}{I^2} \left[(\sigma_1^2 - \sigma_{12}) \left\{ \sigma_1^2 \sigma_2^2 - \sigma_1^2 \sigma_{12} + \sigma_1^2 \sigma_{12} - \sigma_{12}^2 \right\} + (\sigma_1^2 - \sigma_{12}) \left\{ \sigma_1^2 \sigma_2^2 - \sigma_2^2 \sigma_{12} + \sigma_2^2 \sigma_{12} - \sigma_{12}^2 \right\} \right] =$$

$$= \frac{1}{I^2} \left[\sigma_1^2 \sigma_2^2 - \sigma_{12}^2 \right] \left[\sigma_1^2 + \sigma_2^2 - 2\sigma_{12} \right] = \boxed{\frac{\sigma_1^2 \sigma_2^2 - \sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}}$$

$$\sigma_{12} = \rho_{12} \sigma_1 \sigma_2$$

$$\rightarrow \boxed{\sigma^2 = \frac{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12} \sigma_1 \sigma_2} = (1 - \rho_{12}^2) \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} - \frac{2\rho_{12}}{\sigma_1 \sigma_2} \right)^{-1}}$$

Če imamo še $z_3, \sigma_3, \sigma_{13}, \sigma_{23}$:

• izračunamo $\hat{x}_{(12)}, \hat{\sigma}_{(12)}^2$

• izračunamo $\hat{x}_{(123)}, \hat{\sigma}_{(123)}^2$

$$\sigma_{x,w3} = \langle (\hat{x}_{(12)} - x)(z_3 - x) \rangle$$

Varianca povprečja odvisnih keritev

$$\{z_i\}_N, \quad \langle z_i \rangle = \mu$$

$$\langle (z_i - \mu)^2 \rangle = \sigma_i^2$$

$$\langle (z_i - \mu)(z_j - \mu) \rangle = \sigma_{ij}$$

$$\text{Povprečje: } \bar{z} = \frac{1}{N} \sum z_i$$

$$\langle \bar{z} \rangle = \frac{1}{N} \sum \langle z_i \rangle = \frac{1}{N} \cdot N \mu = \mu$$

$$\sigma_{\bar{z}}^2 = \langle (\bar{z} - \mu)^2 \rangle = \frac{1}{N^2} \langle (\sum z_i - N\mu)^2 \rangle = \frac{1}{N^2} \langle (\sum (z_i - \mu))^2 \rangle = \frac{1}{N^2} \langle \sum_i \sum_j (z_i - \mu)(z_j - \mu) \rangle = \frac{1}{N^2} \sum_{i,j} \sigma_{ij} = \frac{1}{N^2} \left(\sum_i \sigma_i^2 + \sum_{i \neq j} \sigma_{ij} \right) = \frac{1}{N^2} \left[\sum_i \sigma_i^2 + 2 \sum_{i < j} \sigma_{ij} \right]$$

Primer:

$$h_1 = 2139m, \sigma_1 = 12m$$

$$h_2 = 2130m, \sigma_2 = 6m$$

$$\sigma_{z2} = 0$$

$$a) \hat{h}_1 = 2135,5m$$

$$\sigma_h^2 = \frac{1}{4}(144 + 36)m^2 = \frac{180}{4}m^2 = 45m^2$$

$$\sigma_h = 6,7m$$

$$b) \hat{h}_1 = h_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} (h_2 - h_1) = 2139m + \frac{144}{180}(-9m) = 2131,8m$$

$$\hat{\sigma}^2 = \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} \right)^{-1} = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \sigma_1^2 - \frac{\sigma_1^4}{\sigma_1^2 + \sigma_2^2} = 28,6m^2$$

$$\hat{\sigma} = 5,4m$$

Združevanje neodvisnih meritev

po N združevanjih:

$$(\hat{x}_N, \hat{\sigma}_N^2)$$

nova meritev (z_{N+1}, σ_{N+1}^2)

$$\hat{x}_{N+1} = \hat{x}_N + \frac{\sigma_N^2}{\sigma_N^2 + \sigma_{N+1}^2} (z_{N+1} - \hat{x}_N)$$

$$\hat{\sigma}_{N+1}^2 = \sigma_N^2 + \sigma_{N+1}^2$$

Kalmanov filter za združevanje neodv. meritev konstante

Uteženo povprečevanje (optimalno)

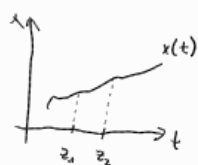
$$\hat{x}_N = \frac{\sum \frac{z_i}{\sigma_i^2}}{\sum \frac{1}{\sigma_i^2}}$$

$$\hat{\sigma}_N^2 = \frac{1}{\sum \frac{1}{\sigma_i^2}}$$

Kalmanov filter za sledenje skalarne spremenljivki (diskretna slika)

$$x = x(t)$$

dinamika: $\dot{x} = A(t)x + C(t)$



$$\frac{x((n+1)T) - x(nT)}{T} = A(nT)x(nT) + C(nT), \quad x_n = x(nT)$$

$$\frac{x_{n+1} - x_n}{T} = A(nT)x_n + C(nT)$$

$$x_{n+1} = (1 + A_n)x_n + C_n, \quad A_n = A(nT)T$$

$$C_n = C(nT)T$$

$$x_{n+1} = \phi_n x_n + C_n + \Gamma_n w_n, \quad \langle w_n w_m \rangle = Q_n \delta_{nm}$$

dinamični
šum
(učinek
motenj)

↓
varianca
din. šuma

0) dinamika

$$x_{n+1} = \phi_n x_n + C_n + \Gamma_n w_n$$

1) napoved

$$\text{ob } nT: \hat{x}_n, \hat{\sigma}_n^2$$

$$\bar{x}_{n+1} = \phi_n \hat{x}_n + C_n$$

$$\bar{\sigma}_{n+1}^2 = \phi_n^2 \hat{\sigma}_n^2 + \Gamma_n^2 Q_n$$

2) Ostrjenje napovedi z meritvijo

$$\text{ob } (n+1)T: z_{n+1}, \sigma_{n+1}^2$$

$$\hat{x}_{n+1} = \bar{x}_{n+1} + \frac{\bar{\sigma}_{n+1}^2}{\bar{\sigma}_{n+1}^2 + \sigma_{n+1}^2} (z_{n+1} - \bar{x}_{n+1})$$

$$\hat{\sigma}_{n+1}^2 = \frac{\bar{\sigma}_{n+1}^2 \sigma_{n+1}^2}{\bar{\sigma}_{n+1}^2 + \sigma_{n+1}^2} = \bar{\sigma}_{n+1}^2 - \frac{\bar{\sigma}_{n+1}^4}{\bar{\sigma}_{n+1}^2 + \sigma_{n+1}^2}$$

novi oznake:

$$\sigma_{n+1}^2 = R_{n+1}$$

$$\bar{\sigma}_{n+1}^2 = P_{n+1}$$

$$\hat{\sigma}_{n+1}^2 = P_{n+1}$$

če ne merimo:

$$z_j$$

$$\hat{x}_{n+1} = \bar{x}_{n+1}$$

$$P_{n+1} = P_{n+1}$$

Primer



Pri vsaki odboju se ohladi δ energije, $\delta = \frac{1}{2}$

izračunaj h_{max} po 5. odboju (h_5)
določi tudi negativnost $\hat{\sigma}_5$

$$x_{n+1} = \phi_n x_n + C_n + \Gamma_n w_n$$

$$x_{n+1} = \phi_n x_n + C_n$$

$$h_1 = \delta h_0 + \delta H$$

$$h_{n+1} = \delta h_n + \delta H$$

$$\rightarrow \phi_n = \delta, C_n = \delta H, \Gamma_n w_n = 0$$

kor. ko merimo

$$\hat{h}_1 = \bar{h}_1 = \delta \hat{h}_0 + \delta H$$

$$\hat{h}_2 = \bar{h}_2 = \delta \hat{h}_1 + \delta H = \delta^2 \hat{h}_0 + \delta^2 H + \delta H$$

$$\hat{h}_3 = \bar{h}_3 = \delta \hat{h}_2 + \delta H = \delta^3 \hat{h}_0 + \delta^3 H + \delta^2 H + \delta H + \delta H$$

$$\hat{h}_4 = \bar{h}_4 = \delta \hat{h}_3 + \delta H = \delta^4 \hat{h}_0 + \delta^4 H + \delta^3 H + \delta^2 H + \delta H + \delta H$$

$$\hat{h}_5 = \bar{h}_5 = \delta \hat{h}_4 + \delta H = \delta^5 \hat{h}_0 + \delta^5 H + \delta^4 H + \delta^3 H + \delta^2 H + \delta H + \delta H$$

$$\hat{x}_n = \sum_{i=0}^n \delta^i = \delta + \delta^2 + \dots + \delta^n = \delta(1 + \delta + \dots + \delta^{n-1}) = \delta(1 - \delta^n) / (1 - \delta)$$

$$\rightarrow \hat{h}_1 = \delta \frac{1 - \delta^n}{1 - \delta}$$

$$\rightarrow \hat{h}_2 = \delta^2 \frac{1 - \delta^n}{1 - \delta} + \delta H$$

$$\hat{h}_5 = H(1 - \delta^5) = \frac{31}{32} H$$

$$P_{n+1} = \phi_n^2 P_n + \Gamma_n^2 Q_n = \delta^2 P_n$$

$$P_5 = \delta^5 P_0 = \delta^5 \sigma_0^2 \rightarrow \hat{\sigma}_5^2 = \delta^5 \sigma_0^2 = \frac{1}{32} \sigma_0^2$$

Primer s šumom:

ista kot prej, razen $\sigma_h \neq 0$

$$H = \bar{H} + \eta, \quad \langle \eta \rangle = 0$$

$$\langle \eta^2 \rangle = \sigma_\eta^2$$

$$h_{n+1} = \delta h_n + \delta(\bar{H} + \eta) = \delta h_n + \delta \bar{H} + \delta \eta$$

$$\rightarrow \phi_n = \delta, \quad c_n = \delta \bar{H}, \quad \Gamma_n \omega_n = \delta \eta$$

$$\hat{h}_{n+1} = \bar{h}_{n+1} = \phi_n h_n + c_n$$

$$\hat{h}_5 = \frac{31}{32} \bar{H}$$

$$P_{n+1} = M_{n+1} = \phi_n^2 P_n + \Gamma_n^2 Q_n$$

$$\langle (\Gamma_n \omega_n)^2 \rangle = \Gamma_n^2 \langle \omega_n^2 \rangle = \Gamma_n^2 Q_n$$

$$\Gamma_n^2 Q_n = \langle (\delta \eta)^2 \rangle = \delta^2 \langle \eta^2 \rangle = \delta^2 \sigma_\eta^2$$

$$\rightarrow P_1 = \delta^2 \sigma_0^2 + \delta^2 \sigma_\eta^2$$

$$P_2 = \delta^2 P_1 + \delta^2 \sigma_\eta^2 = \delta^4 \sigma_0^2 + \delta^4 \sigma_\eta^2 + \delta^2 \sigma_\eta^2$$

$$P_n = \delta^{2n} \sigma_0^2 + \sigma_\eta^2 \sum_{i=0}^{n-1} \delta^{2i}$$

$$h_2 = \delta^2 h_0 + H(\delta + \delta^2) = f(h_0, H)$$

po Gaussu:

$$\hat{\sigma}_2^2 = \delta^4 \sigma_0^2 + (\delta + \delta^2)^2 \sigma_\eta^2$$

$$= \delta^4 \sigma_0^2 + (\delta^2 + \delta^4 + 2\delta^3) \sigma_\eta^2$$

dobimo zlede pride, ker predp., da so vse stopnice enake

Kalmanov filter za sledenje vekt. spr. (diskretna slika)

$$x_1, \dots, x_k \rightarrow \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix}$$

a) Dinamika

$$\vec{x}_{n+1} = \phi_n \vec{x}_n + \vec{c}_n + \Gamma_n \vec{w}_n$$

ϕ_n matrika

$$\vec{x} = \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_k \end{pmatrix}$$

$$\hat{x} = \begin{pmatrix} \hat{x}_1 \\ \vdots \\ \hat{x}_k \end{pmatrix}$$

b) Napoved
ob uT: $\hat{x}_n, P_n$ kovariančna matrika izračunane

$$\vec{x}_{n+1} = \phi_n \hat{x}_n + \vec{c}_n$$

$$M_{n+1} = \langle (\vec{x}_{n+1} - \hat{x}_{n+1})(\vec{x}_{n+1} - \hat{x}_{n+1})^T \rangle$$

$$\vec{x}_{n+1} - \hat{x}_{n+1} = \phi_n \hat{x}_n + \vec{c}_n - (\phi_n \hat{x}_n + \vec{c}_n + \Gamma_n \omega_n) = \phi_n (\hat{x}_n - \vec{x}_n) - \Gamma_n \omega_n$$

$$M_{n+1} = \langle [\phi_n (\hat{x}_n - \vec{x}_n) - \Gamma_n \omega_n] [\dots]^T \rangle$$

$$\text{predp. } \langle \phi_n (\hat{x}_n - \vec{x}_n) \omega_n^T \Gamma_n^T \rangle = 0$$

$$M_{n+1} = \langle \phi_n (\hat{x}_n - \vec{x}_n) (\hat{x}_n - \vec{x}_n)^T \phi_n^T \rangle + \langle \Gamma_n \omega_n \omega_n^T \Gamma_n^T \rangle =$$

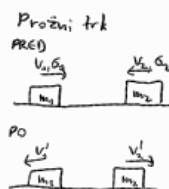
$$M_{n+1} = \phi_n P_n \phi_n^T + \Gamma_n Q_n \Gamma_n^T$$

če ne menimo:

$$P_{n+1} = M_{n+1}$$

$$\hat{x}_{n+1} = \vec{x}_{n+1}$$

Primer



Izračunaj kov. matriko

po popolnoma prožnem trku za vektor

$$\vec{v}' = \begin{pmatrix} v_1' \\ v_2' \end{pmatrix}$$

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \quad P = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix}$$

$$\vec{v}' = \phi \vec{v} + \vec{c} + \Gamma \vec{w}$$

težiščni sistem

$$u_1 = v_1 - v_2$$

$$u_2 = v_1 + v_2$$

$$\text{ohranj. p.: } m_1 u_1 + m_2 u_2 = 0 \rightarrow u_2 = -\frac{m_1}{m_2} u_1$$

$$m_1 u_1 + m_2 u_2 = 0 \rightarrow u_2' = -\frac{m_1}{m_2} u_1'$$

$$\text{okl. T: } \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 u_1'^2 + \frac{1}{2} m_2 u_2'^2$$

$$m_1 u_1^2 + \frac{m_1^2}{m_2} u_1^2 = m_1 u_1'^2 + \frac{m_1^2}{m_2} u_1'^2$$

$$u_2 = u_1'^2$$

$$u_1' = \pm u_1$$

$$u_1' = -u_1$$

$$u_2' = -u_2$$

$$\rightarrow u_1' = -u_1$$

$$v_1' = -v_1 + 2v_2$$

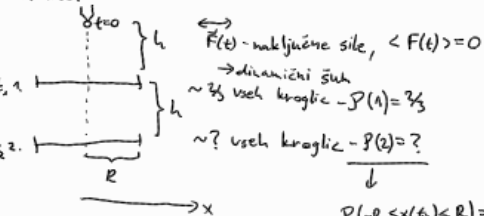
$$v_2' = -v_2 + 2v_1$$

Kalmanov filter za vekt. spr. v zv. sliki

$$\dot{\vec{x}} = A\vec{x} + \vec{c}(t) + \Gamma \vec{w}(t)$$

$$\dot{P} = AP + PA^T + \Gamma Q \Gamma^T - PH^T R^{-1} H P$$

Primer



$$P(-R < x(t_2) < R) = P(x(t_2) < R) - P(x(t_2) < -R) =$$

$$= F\left(\frac{R-0}{\sigma_x(t_2)}\right) - F\left(\frac{-R-0}{\sigma_x(t_2)}\right) = 2F\left(\frac{R}{\sigma_x(t_2)}\right) - 1$$

↑
kumulativna funkcija

$$\dot{\vec{x}} = A\vec{x} + \vec{c}(t) + \Gamma \vec{w}$$

$$\text{iščemo } \sigma_x(t) = \sqrt{P_{11}}$$

$$\begin{pmatrix} \dot{x} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ v \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \omega$$

$$\dot{P} = AP + PA^T + \Gamma Q \Gamma^T$$

$$P(t=0) = \begin{pmatrix} \sigma_0^2 & 0 \\ 0 & 0 \end{pmatrix}$$

$$(AP)^T = P^T A^T = PA^T$$

$$P = \begin{pmatrix} \sigma_x^2 & \sigma_{xv} \\ \sigma_{xv} & \sigma_v^2 \end{pmatrix} = \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix}$$

$$AP = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix} = \begin{pmatrix} P_{21} & P_{22} \\ 0 & 0 \end{pmatrix}$$

$$PA^T = (AP)^T = \begin{pmatrix} P_{21} & 0 \\ P_{22} & 0 \end{pmatrix}$$

$$\Gamma Q \Gamma^T = \begin{pmatrix} \Gamma_1^2 & 0 \\ 0 & 0 \end{pmatrix}$$

↳ VEDNO DIAGONALNA

↳ variiran din. sistem

$$\sigma_x(0) = 0$$

$$\sigma_v(0) = 0$$

• dinamika:

$$\sum F_i = m a_x = m \ddot{x} = F(t)$$

$$\ddot{x} = \frac{F(t)}{m} = a(t) = \omega(t)$$

$$\rightarrow \begin{pmatrix} \dot{x} = v \\ \dot{v} = \omega \end{pmatrix}$$

$$\vec{x} = \begin{pmatrix} x \\ v \end{pmatrix}$$

$$\begin{aligned} \dot{p}_{11} &= 2p_{12} \\ \dot{p}_{12} &= p_{22} \\ \dot{p}_{22} &= Q \end{aligned} \quad \begin{aligned} p_{22}(t) &= Qt + p_{22}(0) \\ p_{12}(t) &= \frac{1}{2}Qt^2 + p_{12}(0) \\ p_{11}(t) &= \frac{1}{3}Qt^3 + p_{11}(0) \end{aligned}$$

$$\rightarrow \frac{1}{3}Qt_1^3 = R^2$$

$$Q = \frac{3R^2}{t_1^3}$$

$$\rightarrow \sigma_x^2(t_2) = R^2 8^{1/2}$$

$$\sigma_x(t_2) = R 8^{1/4}$$

$$\begin{aligned} \sigma_x^2(t_1) &= \frac{1}{3}Qt_1^3 \\ \sigma_x^2(t_2) &= \frac{1}{3}Qt_2^3 \\ P(-R < x(t_2) < R) &= 2/3 \\ P(-\sigma_x(t_1) < x(t_1) < \sigma_x(t_1)) &\approx 68\% \end{aligned}$$

$$\rightarrow \sigma_x^2(t_2) = R^2 \left(\frac{t_2}{t_1}\right)^3$$

$$h = \frac{1}{2}gt_1^2 \rightarrow t_1 = \sqrt{\frac{2h}{g}}$$

$$2h = \frac{1}{2}gt_2^2 \rightarrow t_2 = 2\sqrt{\frac{h}{g}}$$

$$\rightarrow P(-R < x(t_2) < R) = 2F\left(\frac{R}{\sigma_x(t_2)}\right) - 1 = 2F\left(\frac{1}{8^{1/4}}\right) - 1 = 2F(0.724) - 1 = 0.55$$

Primer

po dolgem času je varianca hitrosti enaka 1/3 variance hitr. na začetku

a) zapiši Kalmanov filter za $\hat{x} = [y]$

b) poišči $\sigma_{\hat{y}}^2(\tilde{t})$, $\beta\tilde{t} = 1$

$\tilde{F}_h = -h\beta\tilde{v}$
upoštevaj tudi din. ških

a) dinamika

$$\Sigma \dot{F}_i = m a_{iy} = m \ddot{y}$$

$$m \ddot{y} = m g - F_{\text{vzr}} - m \beta \tilde{v} + F(t)$$

$$\ddot{y} = g_{\text{eff}} - \beta \tilde{v} + \frac{F(t)}{m}$$

$$\dot{\tilde{x}} = \begin{bmatrix} \dot{y} \\ \dot{\tilde{v}} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\beta \end{bmatrix} \begin{bmatrix} y \\ \tilde{v} \end{bmatrix} + \begin{bmatrix} 0 \\ g_{\text{eff}} \end{bmatrix} + \begin{bmatrix} 0 \\ w \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -\beta \end{bmatrix} \quad P = \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix}$$

$$P A^T = (A P)^T = \begin{bmatrix} p_{11} - \beta p_{12} & p_{12} - \beta p_{22} \\ p_{12} - \beta p_{22} & p_{22} - \beta p_{22} \end{bmatrix}$$

$$\Gamma Q \Gamma^T = \begin{bmatrix} 0 & 0 \\ 0 & Q \end{bmatrix}$$

$$\dot{p} = \begin{bmatrix} 2p_{12} & p_{22} - \beta p_{12} \\ p_{12} - \beta p_{22} & Q - 2\beta p_{22} \end{bmatrix}$$

b) iščemo $\sigma_{\hat{y}}^2(\tilde{t}) = p_{22}(\tilde{t})$

$$\dot{p}_{22} = Q - 2\beta p_{22}$$

$$u = Q - 2\beta p_{22}$$

$$\dot{u} = -2\beta u$$

$$u(t) = C e^{-2\beta t}$$

$$u(0) = C$$

$$Q - 2\beta p_{22} = (Q - 2\beta p_{22}(0)) e^{-2\beta t}$$

$$t \rightarrow \infty:$$

$$Q - 2\beta p_{22}(\infty) = 0$$

$$Q = 2\beta p_{22}(\infty)$$

$$\rightarrow \cancel{p}_{22}(\infty) - \cancel{p}_{22}(t) = (\cancel{p}_{22}(\infty) - \cancel{p}_{22}(0)) e^{-2\beta t}$$

$$p_{22}(t) = p_{22}(\infty) - (p_{22}(\infty) - p_{22}(0)) e^{-2\beta t}$$

$$p_{22}(\infty) = \frac{1}{4} p_{22}(0)$$

$$p_{22}(t) = \left[\frac{1}{4} + \frac{3}{4} e^{-2\beta t} \right] p_{22}(0)$$

$$p_{22}(\tilde{t}) = \frac{p_{22}(0)}{4} [1 + 3e^{-2}]$$

Kalmanov filter za nelinearno dinamiko (razširjeni Kalmanov filter)

$$\begin{aligned} \hat{x}_{n+1} &= f(\hat{x}_n) + \Gamma_n^T \tilde{w}_n \\ \hat{z}_{n+1} &= h(\hat{x}_{n+1}) + \tilde{v}_{n+1} \end{aligned}$$

priblišek

$$\hat{x}_{n+1} = f(\hat{x}_n) + J_f(\hat{x}_n) (\hat{x}_n - \hat{x}_n) + O(2) + \Gamma_n^T \tilde{w}_n$$

napoved

$$\hat{x}_{n+1} = f(\hat{x}_n)$$

$$M_{n+1} = J_f(\hat{x}_n) P_n J_f^T(\hat{x}_n) + \Gamma_n Q_n \Gamma_n^T$$

ostrenje

$$\tilde{z}_{n+1} = h(\hat{x}_{n+1}) + \tilde{v}_{n+1} = h(\hat{x}_{n+1}) + J_h(\hat{x}_{n+1}) (\hat{x}_{n+1} - \hat{x}_{n+1}) + O(2) + \tilde{v}_{n+1}$$

$$\hat{x}_{n+1} = \hat{x}_{n+1} + K_{n+1} (\tilde{z}_{n+1} - h(\hat{x}_{n+1}))$$

$$P_{n+1} = M_{n+1}^{-1} + J_h^T(\hat{x}_{n+1}) R_{n+1}^{-1} J_h(\hat{x}_{n+1})$$

$$K_{n+1} = P_{n+1} J_h^T(\hat{x}_{n+1}) R_{n+1}^{-1}$$

Primer:

$$\tilde{z}_{n+1} = \varphi_0 + A \sin(\omega t + \delta) + r_{n+1}$$

$$\tilde{x} = \begin{bmatrix} \varphi_0 \\ A \\ \omega \\ \delta \end{bmatrix}$$

dinamika:

$$\dot{\tilde{x}}_{n+1} = \tilde{x}_{n+1} \Rightarrow \phi_n = I, c_n = 0, \Gamma_n \tilde{w}_n = 0$$

($\varphi_0, A, \omega, \delta$ so konst.)

napoved

$$\tilde{x}_{n+1} = \tilde{x}_n$$

$$M_{n+1} = P_n$$

ostrenje

$$\tilde{z}_{n+1} = h(\hat{x}_{n+1}) + r_{n+1}$$

$$h(\hat{x}_{n+1}) = \varphi_0 + A \sin(\omega t_{n+1} + \delta)$$

$$H_{n+1} = J_h(\hat{x}_{n+1}) =$$

$$= \begin{bmatrix} \frac{\partial h}{\partial \varphi_0} & \frac{\partial h}{\partial A} & \frac{\partial h}{\partial \omega} & \frac{\partial h}{\partial \delta} \end{bmatrix} =$$

$$= \begin{bmatrix} 1 & \sin(\omega t_{n+1} + \delta_{n+1}) & A \omega t_{n+1} \cos(\omega t_{n+1} + \delta_{n+1}) & A \cos(\omega t_{n+1} + \delta_{n+1}) \end{bmatrix}$$